COMPARISON AND LOGICAL OPERATORS

BASICS OF C++

prepared by:

Gyro A. Madrona

Electronics Engineer

GRADING SYSTEM

FINAL GRADE

30%

PRELIM

30%

MIDTERM

40%

ENDTERM

TERM GRADE

50%

WEEKLY

50%

EXAM

100%

LABORATORY

PENALTY

0

ABSENT

3

45-min LATE

7 max

15-min LATE

TOPIC OUTLINE

Relational and Comparison Operators

Logical Operators

RELATIONAL AND COMPARISON OPERATORS

RELATIONAL AND COMPARISON OPERATORS

==	**Equal to**
! =	**Not equal to**
<	**Less than**
>	**Greater than**
<=	**Less than or equal to**
>=	**Greater than or equal to**

Two expression can be compared using relational and equality operators

The result of such operation is either **true** or **false** (i.e. a Boolean value)

RELATIONAL AND COMPARISON OPERATORS

==	**Equal to**
! =	**Not equal to**
<	**Less than**
>	**Greater than**
<=	**Less than or equal to**
>=	**Greater than or equal to**

(7 == 5) evaluates to **false**

(5 > 4) evaluates to **true**

(3 ! = 2) evaluates to **true**

(6 >= 6) evaluates to **true**

(5 < 5) evaluates to **false**

LOGICAL OPERATORS

RELATIONAL AND COMPARISON OPERATORS

!	NOT operator
&&	AND operator
||	OR operator

RELATIONAL AND COMPARISON OPERATORS

!	**NOT operator**
&&	AND operator
||	OR operator

The NOT operator returns the **opposite** Boolean value of the operand

RELATIONAL AND COMPARISON OPERATORS

!	**NOT operator**
&&	AND operator
||	OR operator

! (5 == 5) evaluates to **false**

! (6 <= 4) evaluates to **true**

! true evaluates to **false**

! false evaluates to **true**

RELATIONAL AND COMPARISON OPERATORS

!	NOT operator
&&	**AND operator**
||	OR operator

AND Operator		
A	B	A && B
0	0	0
0	1	0
1	0	0
1	1	1

$(\mathbf{5} == \mathbf{5})$ && $(\mathbf{3} > \mathbf{6})$ evaluates to **false**

RELATIONAL AND COMPARISON OPERATORS

!	NOT operator
&&	AND operator
||	**OR operator**

OR Operator		
A	**B**	**A || B**
0	**0**	**0**
0	**1**	**1**
1	**0**	**1**
1	**1**	**1**

$(\mathbf{5} == \mathbf{5})$ || $(\mathbf{3} > \mathbf{6})$ evaluates to **true**

```
// Grade Calculator

#include <iostream>

#include <string>

using namespace std;

int main( ) {
    // variables
    string name = “complete name”;

    float prelim = 0.1;

    float midterm = 0.1;

    float end_term = 0.1;

    float final_grade = 0.1;
```

```
// prompt for user name
cout<<"Enter your complete name: ";
getline(cin, name);

// prompt the user to enter grades
cout<<"Enter PRELIM grade: ";
cin>>prelim;
cout<<"Enter MIDTERM grade: ";
cin>>midterm;
cout<<"Enter ENDTERM grade: ";
cin>>endterm;
```

```
// calculate final grade
final_grade = (0.3*prelim) + (0.3*midterm) + (0.4*endterm);
cout<<name<<" your final grade is "<<final_grade;
cout<<"Remarks: ";

// determine remarks
if( ) {
        cout<<"Excellent";
}
-------------------------------------------
-------------------------------------------
-------------------------------------------
-------------------------------------------

}
```

9.0-10.0	Excellent
8.0-8.9	Superior
7.0-7.9	Very Good
6.0-6.9	Good
5.0-5.9	Satisfactory
Below 5.0	Failure

LABORATORY